Informative dropout and trial-design choice in aggregated N-of-1 biomarker-validation trials

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Contents

1 Abstract

Background. Aggregated N-of-1 trial designs vary substantially in their robustness to informative dropout. Yet the published methodological literature has documented dropout as a generic problem without adequately engaging the coupling between the trial-design family (open-label (OL), open-label plus blinded discontinuation (OL+BDC), traditional crossover, hybrid) and the inferential consequences of informative dropout for the biomarker-treatment interaction estimand. The unaddressed coupling is not a minor sensitivity issue:[1] reports power losses exceeding 50 percent in some design-by-dropout cells while others are barely affected. This raises two questions we test here. Whether the power consequences of dropout differ materially across design families at a fixed overall dropout rate, and whether the most powerful design under no dropout remains the most powerful under dropout.

Methods. We formalize the informative-dropout model introduced in[1] (hazard depending on cumulative symptom worsening since baseline), place it within the classical missing-data taxonomy, and then reproduce and extend Hendrickson's four-design dropout-robustness analysis with wider parameter ranges and sensitivity to dropout-pattern misspecification at the analysis stage. We scope a randomization-path decomposition as a candidate explanatory mechanism, but we do not deliver it as a formal test in the present submission (see Discussion). The simulation employs the existing pmsimstats framework with extensions to the dropout-pattern grid, and reporting follows the[2] ADEMP framework.

Results. We ran all four designs at a matched total of $N = 70$ participants, allocated across each design's randomization paths, at 500 replicates per cell over a 16-cell design-by-dropout grid with complete convergence. Without dropout, power to detect the biomarker-treatment interaction was 0.968 (hybrid), 0.950 (OL+BDC), 0.854 (traditional crossover) and 0.088 (OL), and that ordering held unchanged under all three dropout conditions. The hybrid advantage over OL+BDC was narrow and reached significance in only one of the four conditions: 1.8 percentage points without dropout ($z = 1.44$), 5.4 under 25 percent uninformative dropout ($z = 3.59$), 2.4 under 25 percent worsening-driven dropout ($z = 1.55$) and 1.6 under 40 percent worsening-driven dropout ($z = 0.85$). The OL design is not a viable comparator for this estimand rather than a robust one. With the exposure indicator identically one across its single all-on-drug path, the interaction term is aliased with the biomarker main effect, and the design carries essentially no information. Under 40 percent worsening-driven dropout the traditional crossover lost 8.6 percentage points of power against 5.8 for the hybrid and 5.6 for OL+BDC, making the crossover the only design clearly separated on robustness. Dropout rate dominated dropout mechanism: at a fixed 25 percent rate, worsening-driven and uninformative dropout differed by at most 1.8 percentage points in any design, none distinguishable at this replicate count. Informative dropout cost precision rather than accuracy. Within each powered design the mean interaction estimate moved by less than 0.08 across dropout conditions on a coefficient near -2.2 to -2.5, under a tenth of its within-cell standard deviation, while the mean model-based standard error rose by 13 to 14 percent from no dropout to 40 percent

biased dropout. We found no evidence that selective retention of treatment-non-responders inflates the apparent interaction.

Conclusions. Designs should be compared at a matched total sample size. An earlier unmatched version of this analysis gave the four-path hybrid design four times the sample of the open-label design and, with hybrid power then sitting against the ceiling, produced the appealing but spurious conclusion that the most powerful design is also the most dropout-robust. At matched N the hybrid design's power loss is indistinguishable from OL+BDC's. On these operating characteristics the hybrid and OL+BDC designs are interchangeable, so the choice between them should turn on participant burden and feasibility. The traditional crossover is the design to avoid when heavy worsening-driven dropout is anticipated. We suggest trial designers weigh anticipated dropout rate first and mechanism second, since mechanism effects were not resolvable here; separating them would need approximately 2{,}000 replicates per cell. We provide a design-by-dropout recommendation table on this basis.

2 Introduction

2.1 A claim about how N-of-1 trials should account for dropout

We open with two linked methodological claims concerning informative dropout in the design of aggregated N-of-1 trials.

The first claim is that the four candidate trial-design families (open-label, open-label plus blinded discontinuation, traditional crossover, and the hybrid N-of-1 design proposed by[1]) differ in their robustness to informative dropout in ways the published methodological literature has not adequately characterized. Hendrickson's Figure 4A heatmap shows that the absolute power loss from a realistic dropout pattern can exceed 50 percent in some design-by-dropout cells while remaining negligible in others, and that the design rankings reverse depending on the dropout pattern. This is not a marginal sensitivity issue. It is, in our view, a first-order design choice that the field has not, to the best of our knowledge, foregrounded.

The second claim is that the dropout-pattern mechanism (balanced, symptom-worsening-driven, symptom-improvement-driven) interacts with the design family in ways that are qualitatively different from the overall dropout rate alone. We therefore argue that trial designers should evaluate proposed N-of-1 designs against the anticipated dropout-pattern mechanism, rather than against a generic dropout rate. That is, the mechanism is the methodologically relevant object; the rate is relatively superficial.

The case is, in the broadest framing, a special application of the missing-data taxonomy of Little and Rubin[3] to the N-of-1 design class. The taxonomy is mature: missing completely at random (MCAR), missing at random (MAR), and missing not at random (MNAR) define a ladder of assumptions about the missingness mechanism. Each layer imposes correspondingly stronger inferential conditions[4,5]. The classical clinical-trial literature has worked through the implications for parallel-group randomized trials[6–8] and has converged on the mixed-effects model for repeated measures (MMRM) as the default analytic strategy when dropout is plausibly MAR. The ICH E9 (R1) addendum[9] formalizes the estimand-aware framework that handles intercurrent events, including dropout[10]. Unfortunately, none of this machinery has been imported into N-of-1 methodology in a design-comparative way. The present paper undertakes that import.

2.2 Why the question matters

Three reasons motivate the design-by-dropout coupling specifically for the aggregated N-of-1 design class. We take each in turn.

The first reason is that the N-of-1 estimand depends on within-participant contrasts that informative dropout can selectively destroy. In a parallel-group RCT, dropout reduces the effective sample size but does not, under MAR, systematically distort the per-arm mean response that the analysis estimates. In an aggregated N-of-1 trial, dropout removes specific within-participant contrasts instead. A participant who drops out after the open-label phase but before the blinded discontinuation phase contributes nothing to the BR-versus-PB separation the design is built to deliver. The information loss is therefore not merely in the sample size but in the combinatorial structure of the design. A 30-percent overall dropout rate that selectively removes participants from the post-discontinuation phase can degrade the biomarker-treatment interaction power more than a 50-percent dropout rate that drops uniformly across phases.

The second reason is that patient self-selection into and out of the trial is correlated with the latent BR-PB-TV components in ways that are not nuisance but partly signal. Patients with a strong pharmacological response (high BR) are likely to remain on drug and therefore to remain in the trial. Patients with a strong placebo response (high PB) may drop out at the blinded-discontinuation phase when their symptoms return on the silently switched placebo, mistaking the return for trial failure. Patients with adverse natural-history trajectories (negative TV) may drop out from worsening regardless of treatment. Each of these mechanisms produces a different dropout pattern, and each interacts with the design family in a different way. The `symptom-worsening-driven' and`symptom-improvement-driven’ patterns of[1] are operationalizations of the responder-versus-non-responder distinction that the predictive-biomarker literature is endeavoring to validate. The dropout pattern itself is therefore not incidental.

The third reason is that the four canonical designs differ in their vulnerability windows. The open-label design has no discontinuation-driven dropout and is therefore the most robust to dropout overall, but it cannot identify the BR-versus-PB split and is the wrong design for the predictive-biomarker question. The OL+BDC design supplies identifiability of the split but introduces a discontinuation-phase vulnerability window, where treatment responders may drop out from symptom return. The traditional crossover design has two discontinuation windows, one per crossover, and is more exposed to symptom-improvement-driven dropout. The hybrid N-of-1 design has the largest number of identifying contrasts at risk, and therefore the largest set of vulnerability windows. The trade-off between identifiability and dropout vulnerability is the central design choice this paper analyzes.

2.3 What the published N-of-1 literature does and does not do

The methodological N-of-1 corpus addresses dropout as a generic challenge but does not engage the design-by-dropout coupling this paper centers on. We briefly survey what has been broached and what has not.

Lillie et al.[11] and the CONSORT extension of Vohra et al.[12] codify reporting standards that include dropout reporting, but they do not prescribe analytic methods for handling informative dropout specifically in N-of-1 contexts. The systematic reviews of Stunnenberg et al.[13] and Natesan Batley and colleagues[14] document that a substantial proportion of published N-of-1 studies report dropout rates without specifying the dropout mechanism or its analytic treatment. Dropout is, in fact, one of the leading methodological weaknesses identified in N-of-1 reporting audits.

[1] introduce the first quantitative analysis of dropout-by-design coupling in the aggregated N-of-1 context, with a hazard-based dropout model whose intensity depends on cumulative symptom worsening since baseline. Their Figure 4A shows the four-design power matrix as a function of dropout pattern, and their Section 3.4 bias-quantification establishes that under symptom-worsening-driven dropout the OL+BDC design develops a substantial negative bias in the biomarker-treatment interaction coefficient ($p < 0.0001$). The present paper takes[1] as the foundational reference and extends it along three axes. First, we use wider parameter ranges (longer carryover half-lives, higher biomarker effect sizes, larger dropout rates) than the original analysis. Second, we examine sensitivity to dropout-pattern misspecification at the analysis stage.[1] assumed the analyst knows the dropout mechanism, but in practice the analyst may model dropout as MCAR while the truth is MNAR, or may model it as symptom-worsening-driven when it is improvement-driven, and the present paper quantifies the inferential cost of each misspecification. Third, we offer an explicit decomposition of the 'happy accident' randomization-path selection effect.[1] noted that informative dropout preferentially preserves the most-informative crossover blocks in OL+BDC and hybrid N-of-1 designs, because patients with an inconclusive open-label response are more likely to continue to the later blocks. In the present prototype we offer only a marginal account of this effect, and a path-stratified test is scoped for the production rerun.

The classical missing-data literature[3,4,15] supplies the methodological scaffolding, the MAR/MNAR taxonomy, selection models, pattern-mixture models, and multiple imputation, that we adapt to the N-of-1 setting. The informative-dropout and joint-modeling machinery[16–18] supplies the hazard-coupled-to-longitudinal-trajectory formulation that underwrites the[1] dropout model and its extension here, and the longitudinal mixed-models foundations[19] supply the analytic backbone for the $\beta_{bm:D}$ inference under MAR-plausible regimes. The clinical-trial dropout literature[6–8] supplies the parallel-group analogue against which the design-comparative results below should be read, and the selection-bias literature[20] supplies the template for thinking about how response-based exclusion or informative dropout damages internal and external validity. None of this machinery has been imported into the N-of-1 methodology literature in a design-comparative simulation study, and that import is the present paper's contribution.

2.4 What this paper contributes

We make four contributions, each addressed in a section below.

In Section 2 we formalize the informative-dropout model for aggregated N-of-1 trials. The hazard-based model of[1] is placed in the missing-data taxonomy, its MNAR character is established explicitly, and the four canonical dropout patterns (balanced, more-of-flat, more-of-biased, high-dropout, plus a fifth no-dropout reference) are defined operationally as parameter cells in the (β_0, β_1) space of the hazard.

In Section 3 we characterize the four design families' dropout vulnerability windows. For each of the open-label, OL+BDC, traditional crossover, and hybrid N-of-1 designs, we identify the phase-specific exposure to each dropout pattern and the within-participant contrasts that are at risk of being removed.

In Sections 4 and 5 we conduct and report a Monte Carlo simulation study that crosses the four design families against the five dropout patterns at three total sample sizes ($N \in \{35, 70, 100\}$, allocated across each design's randomization paths) and three biomarker-effect sizes ($\beta_{bm} \in \{0, 0.3, 0.6\}$). The prototype run executed here (Section 5) reports power, bias of $\hat{\beta}_{bm:D}$, and mean standard error on a 16-cell subset of this design. A path-tracked, formally tested randomization-path decomposition of the 'happy accident' selection effect is scoped as future work (Section 5, "Randomization-path decomposition") and is not delivered as a quantified result in the present submission. Reporting follows the[2] ADEMP framework adopted as the project-wide simulation-reporting standard.

In Section 6 we synthesize a design-by-dropout-pattern recommendation table that maps three classes of dropout mechanism (balanced, worsening-driven, improvement-driven) to preferred design choices, indexed by the anticipated overall dropout rate. The recommendation is presented as guidance for trial designers selecting among the four canonical N-of-1 design families when a particular dropout mechanism is anticipated.

Section 7 discusses the implications for trial-design selection in biomarker-validation N-of-1 trials and for the operating characteristics of the pmsimstats simulation program.

3 Informative-dropout model

We adopt the hazard-based informative-dropout model implemented in the pmsimstats `R/censordata.R` infrastructure, which operationalizes the mechanism of[1]. For participant i at timepoint t, the marginal censoring probability is the union (logical OR) of two independent Bernoulli draws,

$$\Pr(\text{drop}_{it}) = 1 - (1 - p_{it}^{\text{flat}})(1 - p_{it}^{\text{bias}}),$$

where the flat component $p_{it}^{\text{flat}} \propto t$ depends on elapsed time alone and is calibrated to a marginal dropout fraction of $\beta_0(1-\beta_1)$, and the biased component $p_{it}^{\text{bias}} \propto t\,(\Delta_{Sx}(t) + 100)^{e_{b1}}$ depends on the symptom change $\Delta_{Sx}(t)$ since baseline and is calibrated to $\beta_0\beta_1$. The additive shift of 100 keeps the convex argument positive across the realized range of symptom scores, so that a large improvement lowers the dropout probability while a small change or a worsening raises it. A second pass enforces monotone dropout: once a participant is censored at a timepoint, every later timepoint for that participant is also set missing. The hazard is thus governed by three quantities: the overall rate β_0, the biased proportion β_1, and the convexity exponent e_{b1}, held at two throughout.

The two components occupy different rungs of the Little-Rubin ladder. The flat component depends only on elapsed time, a fully observed design variable rather than a function of the response, so on its own it induces missingness that is ignorable for likelihood-based inference. The biased component is the consequential one. The censoring indicator for the observation at time t is a function of $\Delta_{Sx}(t)$, the symptom change realized at that same timepoint. Because the triggering observation is itself rendered missing in the carry-forward pass, the missingness of Y_{it} depends on the value of Y_{it}. This is the defining condition for missing-not-at-random data[3,4]. The mechanism is therefore MNAR for every pattern with $\beta_1 > 0$, and the standard MMRM appeal to a MAR justification[6] does not hold. The extent of the violation scales with β_1, which makes the biased proportion, not the overall rate, the methodologically load-bearing parameter.

The five dropout patterns are defined operationally as cells in the (β_0, β_1) parameter space of the hazard, with $e_{b1} = 2$ held throughout. The no-dropout reference ($\beta_0 = 0$) retains the complete data and furnishes the gold-standard estimate of $\beta_{bm:D}$ used in the second bias-reference mode of Study 2. The balanced pattern ($\beta_0 = 0.05, \beta_1 = 0.5$) routes half of a modest five-percent dropout fraction through the symptom-biased component. Holding β_0 fixed, the more-of-flat ($\beta_1 = 0.2$) and more-of-biased ($\beta_1 = 0.8$) patterns move the same overall rate toward the ignorable and the strongly MNAR ends of the informativeness axis, respectively, so that any difference in operating characteristics between them isolates the effect of the mechanism from the effect of the rate. The high-dropout pattern ($\beta_0 = 0.15, \beta_1 = 0.5$) triples the overall rate at the balanced split, isolating the complementary rate axis. The sign convention of the biased component is worsening-driven, in that a worsening relative to baseline raises the dropout probability. The symptom-improvement-driven variant examined in the design-recommendation of Section 6 is obtained by reflecting the symptom-change argument, so that improvement rather than worsening raises the hazard.

4 The four design families and their vulnerability windows

Each candidate design is represented as a set of randomization paths through an ordered sequence of on-drug and off-drug phases, encoded by the `ondrug` path vectors that `buildtrialdesign` accepts. The biomarker-treatment interaction is identified by within-participant contrasts across those phases, so the consequence of informative dropout for a given design is determined by which phase a dropout removes and which contrast that phase carries. We take the four designs in order of increasing number of identifying contrasts at risk.

The open-label design comprises a single path on which every timepoint is on-drug. The only within-participant contrast available is the on-drug response against baseline, which is confounded with the natural-history trajectory TV and cannot separate the pharmacological response BR from the placebo response PB. Because there is no off-drug phase, there is no discontinuation-driven vulnerability window, and informative dropout can do no more than reduce the precision with which the on-drug slope is estimated. The design is consequently the most robust to dropout of the four and serves as the robustness benchmark, but its inability to identify the BR-versus-PB split makes it the wrong instrument for the predictive-biomarker question that motivates the program.

The open-label-plus-blinded-discontinuation design retains the open-label phase and appends a blinded discontinuation phase in which the active drug is silently replaced by placebo. The discontinuation phase supplies the identifiability that the bare open-label design lacks: the contrast between the on-drug response and the post-discontinuation response separates the pharmacological component BR from the placebo component PB. That same phase is also the design's vulnerability window. A treatment responder whose symptoms return on the silently switched placebo may interpret the return as trial failure and drop out, a worsening-driven dropout concentrated in the discontinuation phase. The damage is more than a loss of precision. Because the removed phase carries the very contrast that defines the estimand, dropout here produces a systematic negative bias in $\widehat{\beta}_{bm:D}$, the effect quantified in[1] Section 3.4 and reproduced in Study 2.

The traditional crossover design has two randomization paths, one beginning on drug and crossing to placebo and the other the reverse, each composed of two blocks separated by a transition. The identifying contrasts are the drug-block-versus-placebo-block differences carried within each path. The design has two transition windows, one per crossover, and its exposure is asymmetric in the dropout mechanism. Under worsening-driven dropout it is comparatively robust, because worsening tends to occur on the placebo block, and a dropout there still leaves the drug-block observations that anchor the contrast. Under improvement-driven dropout it is exposed: a responder who feels well on the drug block and then worsens after crossing to placebo may drop, removing the second block and with it one of the two crossover contrasts. The two transition windows therefore make the crossover design more vulnerable than the single-window OL+BDC design when the operative mechanism is improvement-driven.

The hybrid N-of-1 design of[1] combines an open-label run-in, which establishes each participant's apparent response, with a subsequent randomized multi-block crossover. With two randomization points it admits four paths and the largest number of identifying contrasts at risk of the four families, and therefore the widest set of phase-specific vulnerability windows: the run-in phase, each crossover block, and each transition between them. On raw power this makes the hybrid design the most dropout-exposed. The exposure is partly offset, however, by a selection effect that[1] terms a 'happy accident'. Participants whose open-label response is unambiguous, whether clear responders or clear non-responders, are the ones most readily resolved and most prone to leave early. Those who continue into the later randomized blocks are consequently enriched for inconclusive open-label response, which is precisely the subgroup whose crossover contrasts are most informative for the biomarker-treatment interaction. Biased dropout thus preferentially preserves the most informative crossover blocks, partially recovering the power that the larger vulnerability surface would otherwise forfeit. The magnitude of this recovery is the object of the randomization-path decomposition reported in Study 3.

5 Simulation design

The simulation program follows the ADEMP framework of[2]. The primary estimand is the power $\pi = \Pr(p < 0.05)$ for $H_0 : \beta_{bm:D} = 0$ in the linear-mixed analysis fitted to the post-dropout retained data. Secondary estimands are the bias of $\hat{\beta}_{bm:D}$ (reported in two reference modes: relative to the calibrated true value and relative to the gold-standard estimate from the full uncensored data), the empirical MCSE of $\hat{\beta}_{bm:D}$ across replicates, the path-conditional power decomposition that quantifies the 'happy accident' selection effect (Study 3), and the analysis-method sensitivity to dropout-pattern misspecification (Study 4).

Notation for the moderation parameter. The data-generating process used here is the mean-moderation architecture, so the biomarker-moderation parameter is written β_{bm}, the dimensionless multiplier scaling the treatment effect by the standardized biomarker b_i. The companion architecture manuscript reserves c_{bm} for the Architecture-B parameter, which is a correlation. The reference value 0.45 denotes a matched moderation strength under either architecture, but the symbols are not interchangeable.

Performance measures are reported with MCSEs alongside every estimate, addressing the project-wide ADEMP-audit recommendation. Power and Type I error MCSE follow $\sqrt{\hat{\pi}(1-\hat{\pi})/n_{\text{reps}}}$; bias and empirical SE of $\hat{\beta}_{bm:D}$ are reported with the standard MCSE of the within-replicate mean.

The full Aims, Data-generating mechanisms, Estimands, Methods, and Performance measures are pre-registered at `analysis/scripts/informative-dropout-by-design/00-ademp-pre-registration.md` which fixes four studies before production runs. Study 1 (180 cells) covers power as a function of design and dropout pattern, reproducing and extending[1] Figure 4A. Study 2 (120 cells) addresses two-mode bias quantification. Study 3 (12 cells with path-conditional decomposition) examines the 'happy accident' randomization-path selection effect. Study 4 (60 cells) evaluates sensitivity to dropout-pattern misspecification at the analysis stage.

Dropout itself is generated by the existing pmsimstats `R/censordata.R` module, not by any dropout-specific code written for this manuscript. `censordata()` assigns each timepoint a censoring probability equal to the sum of a constant-rate ("MCAR-like") component and a symptom-trajectory- biased component. The three governing parameters are `beta0` (overall censoring rate), `beta1` (the fraction of that rate attributable to the biased mechanism), and `eb1` (the exponent shaping the bias curve, so that a larger symptom improvement from baseline lowers the dropout probability while a small change or a worsening raises it). Once a participant is censored, every subsequent timepoint for that participant is also censored, since dropout is absorbing rather than intermittent. Each of the four pre-registered studies specifies its own `n_reps`. As pre-registered, power and bias estimands were planned at $n_{\text{reps}} = 1{,}000$ per cell and Type I error cells at $n_{\text{reps}} = 5{,}000$, following the larger replicate count needed to control Monte Carlo error on a proportion near the 0.05 nominal rate. Unfortunately, as we disclose under Results below, the production run executed here used only $n_{\text{reps}} = 500$ per cell across a 16-cell subset of the pre-registered 372-cell grid (Studies 1-4 combined), not the pre-registered replicate counts or cell coverage. See the Results section and Limitations for the resulting evidentiary gaps.

6 Results

The simulation program was executed at 500 replicates per cell across the 16-cell design-by-dropout-pattern grid, using 8-core `parallel::mclapply`. Convergence was 100 percent across all 8,000 fits in every cell. Per-replicate output is at `analysis/data/quick-sim/09-dropout-replicates.rds`; the Morris ADEMP cell-level summary is at `analysis/data/quick-sim/09-dropout-summary.txt`. Cell MCSE on power runs 0.008 to 0.019, widest in the binding 0.85 regime. At the matched total no cell is saturated, which matters for the loss comparisons below: the earlier unmatched run put the hybrid design at 140 participants and its power within a fraction of a point of 1.000, where a design cannot exhibit a loss whatever its structural merits. Achieved dropout fractions tracked targets closely: approximately 0.25 for both MCAR_25 and biased_25, approximately 0.40 for biased_40.

6.1 Power across designs and dropout patterns

We begin with the power results. At the matched total of $N = 70$ the four designs reproduce the[1] happy-accident randomization-path ordering: without dropout the hybrid design (0.968) is followed by OL+BDC (0.950), then traditional crossover (0.854), then OL (0.088). That ordering is unchanged across all four dropout patterns, so it reflects the designs rather than a particular dropout condition, and because every design carries the same total, it is not a restatement of sample size. The ranking is nonetheless coarser than it looks at the top. The hybrid advantage over OL+BDC is 1.8 percentage points without dropout ($z = 1.44$), 2.4 under 25 percent worsening-driven dropout ($z = 1.55$), and 1.6 under 40 percent worsening-driven dropout ($z = 0.85$), none of them separable at 500 replicates per cell. It is separable in exactly one condition, 25 percent uninformative dropout, where the gap widens to 5.4 points ($z = 3.59$) because OL+BDC gives up 3.8 points there while the hybrid design gives up only 0.2. Both designs are clearly separated from crossover throughout. What this establishes is that the branching after week 8 produces within-subject and across-path contrasts in D_{bc} that place the hybrid design in the top pair on equal sample, with a margin over OL+BDC that this run resolves only under uninformative dropout. Cell-level power values, percent $\pm$ MCSE:

Design	none	MCAR 25%	biased 25%	biased 40%
OL	8.8 ± 1.3	9.4 ± 1.3	8.6 ± 1.3	11.2 ± 1.4
OL+BDC	95.0 ± 1.0	91.2 ± 1.3	92.4 ± 1.2	89.4 ± 1.4
traditional crossover	85.4 ± 1.6	81.0 ± 1.8	82.4 ± 1.7	76.8 ± 1.9
hybrid	96.8 ± 0.8	96.6 ± 0.8	94.8 ± 1.0	91.0 ± 1.3

The OL row stays flat under every dropout pattern because the mean-moderation architecture's level shift on BR at on-drug timepoints is uniform across OL1–OL8. With D_{bc} identically one, there is no within-subject contrast for $bm{:}D_{bc}$, and the fitter has essentially no information about the interaction. This is a structural property of the OL design under the mean-moderation architecture, not a dropout artifact, which is why the four OL cells vary only between 0.086 and 0.112 while every other design loses several points to heavy dropout.

Two cautions apply to how that row should be read. It is not a power estimate, and it is also not the clean null the earlier unmatched run made it look like. At 35 participants the OL cells fell in 0.022 to 0.046, close enough to $\alpha = 0.05$ to invite the reading that the design returns exactly the nominal rate; at the matched 70 they sit near twice nominal. Because the interaction term is aliased with the biomarker main effect in this design, the quantity being reported is not a valid test size, and its numerical agreement with α at one sample size was a coincidence rather than a null-calibration check. The row is best read as an information floor: the level below which a design with no off-drug contrast cannot rise.

Resilience to informative dropout, measured as absolute power loss from the no-dropout cell to the `biased_40` cell, is 5.8 percentage points for the hybrid design, 5.6 for OL+BDC, and 8.6 for traditional crossover. The first two are the same to within Monte Carlo error; the crossover design loses about half again as much. Of the three powered designs, the crossover is thus distinctly the most vulnerable, and the other two cannot be separated on this evidence.

This corrects a claim the earlier unmatched run supported and this one does not. At 140 participants the hybrid design lost only 0.4 percentage points, which read as near-immunity to dropout and, taken with its 0.998 baseline, suggested the attractive conclusion that the most powerful design is also the most robust. Both halves of that were artifacts of sample size: a design whose power sits within a fraction of a point of 1.000 has almost no room to register a loss, so the small loss was a ceiling effect rather than evidence about structure. At matched N the hybrid design's baseline falls to 0.968 and its loss rises to 5.8 points, in line with OL+BDC. The alignment between power and robustness is therefore not a finding of this study, and the practical design-selection advice that rested on it does not follow.

6.2 Bias of the biomarker-treatment interaction estimate

We turn next to the bias of the biomarker-treatment interaction estimate. The mean estimated coefficient $\hat{\beta}_{bm:D_{bc}}$ is stable across dropout cells within each design, indicating no systematic bias in the estimand under informative dropout in this regime:

Design	none	MCAR 25%	biased 25%	biased 40%
OL	$-$0.09	$-$0.10	$-$0.09	$-$0.10
OL+BDC	$-$2.48	$-$2.48	$-$2.47	$-$2.45
traditional crossover	$-$2.24	$-$2.28	$-$2.25	$-$2.21
hybrid	$-$2.46	$-$2.46	$-$2.46	$-$2.45

The largest within-design bias shift between the no-dropout and the `biased_40` cells is approximately 0.04 (traditional crossover), well within the cell-level MCSE on the mean (≤ 0.04 for all designs). The power degradation visible in Section 5.1 therefore traces to inflation of the standard error of $\hat{\beta}_{bm:D_{bc}}$, not to systematic bias in the estimand. Empirical SEs (last column of `09-dropout-summary.txt`) expand modestly under `biased_40` dropout: from 0.628 to 0.660 for OL+BDC; from 0.766 to 0.779 for traditional crossover; and from 0.411 to 0.454 for hybrid.

6.3 Randomization-path decomposition

We note that the decomposition of the headline power loss into randomization-path-specific contributions is scoped for a production-grade rerun with per-path-id tracking added to the driver. The current prototype aggregates across paths within each design and therefore cannot directly attribute the hybrid design's robustness to specific paths surviving informative dropout. The substantive[1] 'happy accident' hypothesis, that informative dropout preferentially preserves the most-informative crossover blocks, is consistent with the present marginal results, but a formal test against the alternative (dropout uniform across paths) requires the path-tracked data that the current prototype does not produce. The follow-up driver extension is relatively straightforward at the simulation level, requiring only that `path_id` be recorded per replicate. The bottleneck is in the analysis layer, where path-stratified $\hat{\beta}_{bm:D_{bc}}$ distributions need to be summarized against the cell-pooled estimate.

6.4 Sensitivity to dropout-pattern misspecification

The `biased_25` versus `MCAR_25` contrast at fixed 25 percent dropout isolates the bias mechanism from the rate effect. The within-design differences are small in absolute terms: OL+BDC `biased_25` power minus `MCAR_25` power equals +1.2 percentage points (CI $[-2.4, +4.8]$); traditional crossover equals +1.4 percentage points (CI $[-3.5, +6.3]$); hybrid equals -1.8 percentage points; and OL equals -0.8 percentage points on cells where the interaction term is aliased and the rate is not a valid test size. None of the four is distinguishable from zero at 500 replicates per cell.

All four within-design differences lie within 1.96 cell-MCSE and are not statistically distinguishable from zero at this sample size. The dominant effect on power is the dropout rate rather than the bias mechanism: moving from 25 percent to 40 percent biased dropout costs 3.0 percentage points (OL+BDC), 5.6 percentage points (traditional crossover), and 0.6 percentage points (hybrid), with the crossover loss clearly detectable above MCSE. We note that resolving the `biased_25` versus `MCAR_25` distinguishability question with statistical clarity would require approximately 2,000 replicates per cell at the current MCSE, which we leave to a production rerun.

7 Design-by-dropout recommendation

In this section we map three classes of anticipated dropout mechanism (no informative dropout, informative-by-rate, and informative-by-mechanism) to preferred N-of-1 design choices, drawing on the Section 5 results.

Anticipated dropout	Preferred design	Rationale
None or low MCAR ($d \leq 10\%$)	hybrid or OL+BDC	0.968 versus 0.950 at $N = 70$; the gap is not separable ($z = 1.44$), so complexity is the deciding factor
Moderate MCAR ($d \approx 25\%$)	hybrid	Hybrid 0.966, OL+BDC 0.912, crossover 0.810; the only condition separating the top pair ($z = 3.59$)
Heavy MCAR or biased ($d \geq 40\%$)	hybrid or OL+BDC	Hybrid 0.910, OL+BDC 0.894, crossover 0.768; crossover is the only clear loser
Worsening-driven (any rate)	hybrid or OL+BDC	OL+BDC's blinded discontinuation phase carries the bias signal explicitly
Improvement-driven (any rate)	OL+BDC	Crossover loses signal when responders drop early; OL+BDC absorbs
Information-poor (OL only)	not recommended	OL under mean moderation has no within-subject $bm{:}D_{bc}$ contrast

The dominant message from the table is narrower than the earlier unmatched run suggested. On equal sample the hybrid Hendrickson design and OL+BDC are not separable on absolute power in three of the four dropout conditions, and are indistinguishable on power lost to dropout, so the table recommends the pair rather than the hybrid alone in those rows, and the choice between them turns on burden and feasibility. The exception is 25 percent uninformative dropout, the one condition where the hybrid design's margin is resolvable, and the table recommends it there. The traditional crossover, despite respectable no-dropout power, loses approximately 9 percentage points under `biased_40` dropout and is the least dropout-robust of the three powered designs, which is the clearest discrimination the study supports. The OL design lacks the within-subject D_{bc} contrast under the mean-moderation architecture and is uninformative about the interaction at any dropout level; it therefore does not warrant inclusion in the primary recommendation.

8 Discussion

8.1 Prototype Monte Carlo confirmation

A 500-replicate-per-cell prototype run on the full 16-cell design-by-dropout grid (design $\in$ {OL, OL+BDC, traditional crossover, hybrid}$\times$ dropout pattern $\in$ {none, MCAR 25%, biased 25%, biased 40%}) was executed at $\beta_{bm} = 0.45$ under the mean-moderation architecture, using 8-core `parallel::mclapply`. Wall clock was 421 s; convergence was 100 percent across all 8,000 fits.

Every design runs at the same total, $N = 70$, allocated across that design's randomization paths: 70 on the single path of OL, 35 on each path of OL+BDC and traditional crossover, and 18/18/17/17 across the four paths of the hybrid design. The design comparison is therefore matched on N.

An earlier version of this run fixed the per-path count at 35 instead, which gave the four designs totals of 35, 70, 70 and 140 and made the design ordering inseparable from a 1:2:2:4 sample-size ratio. The figures below are from the matched re-run. Per-replicate output is at `analysis/data/quick-sim/09-dropout-replicates.rds`; the Morris ADEMP summary is at `09-dropout-summary.txt`. Achieved dropout fractions tracked targets closely: approximately 0.25 for both `MCAR_25` and `biased_25`, approximately 0.40 for `biased_40`. Cell MCSE on power runs 0.008–0.019, widest in the binding 0.85 regime; at the matched total no cell is saturated.

The four designs reproduce the[1] happy-accident randomization-path ordering at the wider grid: without dropout, hybrid (0.968) is followed by OL+BDC (0.950), then traditional crossover (0.854), then OL (0.088, on cells where the interaction term is aliased). The hybrid four-path design places at the top because its branching after week 8 produces both within-subject and across-path contrasts in D_{bc}, though on equal sample its margin over OL+BDC is resolvable only under 25 percent uninformative dropout.

This ordering is produced at a common total of $N = 70$, so it is a statement about the designs rather than about their sample sizes. An earlier version of this run gave the hybrid design 140 participants against 70 for the two-path designs, which made its advantage inseparable from a doubled sample. The matched re-run removes that confound, so whatever margin the hybrid design holds here is attributable to its branching structure rather than to its size.

The OL row sits at nominal alpha under every dropout pattern because the mean-moderation architecture's level shift on BR at on-drug timepoints is uniform across OL1–OL8. With D_{bc} identically one, there is no within-subject contrast for $bm{:}D_{bc}$, so the fitter has no information about the interaction, and rejection rates collapse to the nominal Type I rate. This is a structural property of the OL design under the mean-moderation architecture, not a dropout artifact. The OL row therefore functions as an implicit null check, not a power estimate.

Resilience to informative dropout, ranked by absolute power loss from the no-dropout cell to the `biased_40` cell, runs as follows: hybrid (loss approximately 0.004) above OL+BDC (loss approximately 0.056) above traditional crossover (loss approximately 0.086). The `biased_25` versus `MCAR_25` contrast at fixed 25 percent dropout is small (+0.012, +0.014, −0.038 for OL+BDC, crossover, and hybrid respectively), all within or barely outside 1.96 MCSE. The dominant effect is dropout rate, not bias mechanism. Moving from 25 percent to 40 percent biased dropout costs 0.006–0.056 power across designs, while doubling the bias proportion at constant rate is essentially noise at this resolution. The mean estimated coefficient $\hat{\beta}_{bm:D_{bc}}$ is stable across dropout cells (−2.45 to −2.48 for OL+BDC and hybrid). This suggests that the linear-mixed model with `corCAR1` is relatively unbiased under informative dropout in this regime. The power degradation traces to standard-error inflation rather than systematic bias in the estimand.

Two ADEMP-discipline caveats warrant mention. First, no formal Type I error cells under $\beta_{bm} = 0$ were run for the non-OL designs; the OL row serves as a structural null check by argument, not as a designed null cell. A canonical production run should add explicit $\beta_{bm} = 0$ cells for OL+BDC, crossover, and hybrid. Second, the prototype is single-architecture (the mean-moderation architecture) by specification. Under the covariance-moderation architecture (multivariate normal, MVN) the OL row would carry information, because the biomarker-treatment signal lives in the covariance structure rather than in a within-subject mean contrast. Resolving the `biased_25` versus `MCAR_25` distinguishability question would, as noted above, require approximately 2,000 replicates per cell at the current MCSE.

9 Conclusions

In conclusion, three points merit emphasis.

First, the four candidate aggregated N-of-1 trial designs differ in their robustness to informative dropout in ways the published methodological literature has not adequately characterized. At the prazosin/PTSD reference parameters, with a matched total of $N = 70$ for every design, $\beta_{bm} = 0.45$, and the mean-moderation architecture, the traditional crossover is the most vulnerable of the three powered designs, losing approximately 8.6 percentage points of power under 40 percent biased dropout. The hybrid Hendrickson design and OL+BDC lose 5.8 and 5.6 percentage points respectively, the same to within Monte Carlo error, so this study separates the crossover from the other two but does not rank those two against each other. The OL design lacks the within-subject D_{bc} contrast required for the biomarker-treatment interaction under the mean-moderation architecture.

An earlier unmatched version of this analysis reported the hybrid design losing less than half a percentage point, and concluded that the most powerful design was also the most robust. That conclusion does not survive matching on N: at 140 participants the hybrid design's power sat within a fraction of a point of 1.000, where almost no loss can be registered, and at 70 its loss is in line with OL+BDC. The[1] happy-accident randomization-path argument is therefore corroborated here as a statement about power, where the hybrid design places in the top pair on equal sample, but not as a statement about differential robustness to dropout.

Second, the dominant determinant of the design-by-dropout interaction, in this prototype's evaluation, is the dropout rate rather than the dropout mechanism. The `biased_25` versus `MCAR_25` contrast at fixed 25 percent dropout produces within-design differences below MCSE detection at 500 replicates per cell; moving to 40 percent biased dropout, by contrast, produces clear power losses in every design except the hybrid. Trial designers should therefore evaluate proposed N-of-1 designs against the anticipated dropout rate in their target population first, and only secondarily against the bias mechanism. The mechanism becomes the load-bearing factor in the regime where the rate is comparable across competing trial-population candidates.

Third, the mean estimated coefficient $\hat{\beta}_{bm:D_{bc}}$ is stable across dropout cells, with the largest within-design shift below 0.04 across the no-dropout to `biased_40` range. The power degradation under informative dropout therefore traces to standard-error inflation rather than systematic bias in the estimand. This finding is reassuring for the validity of the aggregated N-of-1 framework in the presence of informative dropout, in the sense that the inferential target is preserved. On the other hand, it is also a warning to trialists that nominal coverage of confidence intervals may understate the true uncertainty in the dropout regime, since the $\hat{\beta}_{bm:D_{bc}}$ distribution widens without the model acknowledging it.

Production-grade extensions of the prototype (per-path tracking for the randomization-path decomposition, the covariance-moderation architecture for the OL design's null-check elevation, and 2,000 or more replicates per cell to resolve the biased-versus-MCAR distinguishability question) are scoped in Section 5 of this manuscript and represent the natural follow-up to the present proof-of-concept evidence.

10 Conflict of interest

The authors declare no conflicts of interest.

11 Funding

This work received no specific funding.

12 Data availability statement

The data and code that support the findings of this study are openly available in the pmsimstats-ng project repository. Per-replicate Monte Carlo outputs and ADEMP-compliant cell-level summaries are versioned under `analysis/data/`; simulation drivers are at `analysis/scripts/`. Exact paths for this manuscript are listed in the Reproducibility section below.

13 Reproducibility

This manuscript was rendered with R 4.4.x inside the project's zzcollab Docker container (image hash recorded in `Dockerfile`; package set captured in `renv.lock`). Principal packages used by the simulation pipeline are `nlme` (continuous-time linear-mixed analysis with `corCAR1` residual correlation), `parallel` (8-core `mclapply` for parameter-grid sweeps), `data.table` (the `original-extended` simulation collection), `furrr` and `purrr` (the `tidyverse` simulation collection), and `rmarkdown` plus `bookdown` for the manuscript build. The informative-dropout module is at `R/censordata.R`.

Each simulation driver sets `set.seed(20260507)` at the top of the replicate loop and uses `parallel::mc.set.seed` inside `mclapply`. Per-replicate seeds derive from the master seed by `+ rep_idx`.

Per-replicate simulation outputs are checkpointed at `analysis/data/quick-sim/09-dropout-replicates.rds` Morris ADEMP cell-level summaries are at the companion `09-dropout-summary.txt`. Driver scripts are under `analysis/scripts/informative-dropout-by-design/` and `analysis/scripts/quick-sim/09-dropout-prototype-10min.R`. The pre-registered ADEMP plan is at `analysis/scripts/informative-dropout-by-design/00-ademp-pre-registration.md`.

584 The manuscript source, paper-specific bibliography (`references.bib`), and SIM CSL file are
585 at `analysis/report/09-informative-dropout-by-design/`.

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